# “Intersex” in Our Project Means Something Different

*A community clarification from the FINS team*

## YOU MAY HAVE HEARD THIS WORD...

### What does “intersex” mean in FINS?

Our project uses the word intersex in a strictly biological and ecological sense — to describe fish whose sex organs are being disrupted by chemical pollution in water. This has nothing to do with human gender identity, human sexuality, or any social debate.

## NOT WHAT WE MEAN

✕ Human gender identity, LGBTQ+ identities, or debates about sex and gender in people.

## WHAT WE MEAN

✓ Male fish developing female reproductive tissue due to estrogen pollution in rivers — a measurable biological problem.

## THE SCIENCE, SIMPLY

### What is actually happening to the fish?

#### The source: chemical pollution

Hormones from medicines, contraceptive pills, pesticides, and industrial waste enter rivers through sewage. India’s wastewater treatment plants cannot remove these chemicals. They pass straight through and into the river.

#### What happens to male fish

When male fish absorb these estrogen-like chemicals, their reproductive organs begin developing female characteristics — egg tissue appears inside the testes. Fishermen see more and more females in the net; the males have become functionally female.

#### Why fish populations crash

Affected males produce less sperm and fertilize fewer eggs. Over time, fewer fish are born each season. This is not a natural change — it is pollution-driven reproductive failure.

**In plain terms:** The river is contaminated by chemicals that mimic estrogen. Fish are chemically confused about their own biology. This is an environmental crisis — not a moral or social one.

## WHY IT MATTERS TO YOU

#### The fishermen’s story

Over 28 million people in India depend on fishing for their livelihood. Many fishermen along the Ganga, Godavari, and Cauvery have noticed declining catches and skewed sex ratios in their nets — without knowing why.

FINS is our attempt to build a solution — a biological sensor that detects estrogen pollution and an enzyme that breaks it down — so that India’s rivers can be restored to health, and fishermen’s livelihoods protected.

> “Intersex” in our project is a medical-ecological term, the same way “fracture” describes a broken bone — factual, clinical, and not political.

## OUR COMMITMENT

### We respect all communities

We are aware that the word “intersex” carries different meanings in different conversations. We chose it because it is the internationally recognised scientific term for this biological condition in fish, used in published research worldwide.

We do not use this word to make any statement about human identity. Our work is rooted entirely in ecology, water chemistry, and public health.

If you have questions, concerns, or want to share what you have observed in your local water bodies, please reach out — your knowledge as a community member matters to our research.